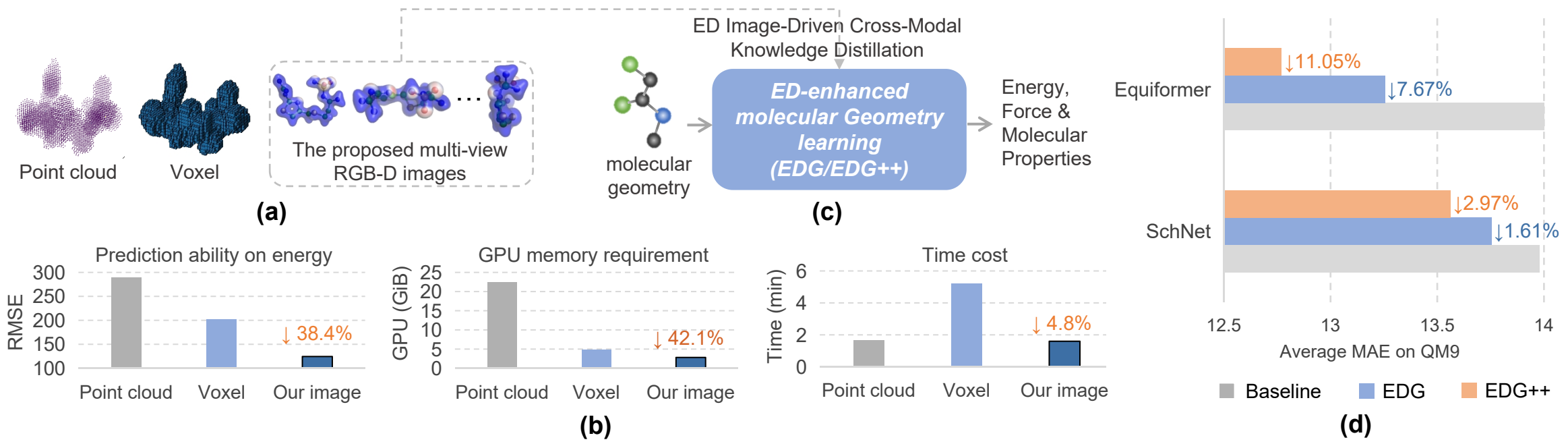
Point cloud
Voxel
The proposed multi-view RGB-D images
(a)
ED Image-Driven Cross-Modal Knowledge Distillation
molecular geometry
ED-enhanced molecular Geometry learning (EDG/EDG++)
Energy, Force & Molecular Properties
(c)
Prediction ability on energy
RMSE
300
250
200
150
100
↓ 38.4%
Point cloud
Voxel
Our image
GPU memory requirement
GPU (GiB)
25
20
15
10
5
0
↓ 42.1%
Point cloud
Voxel
Our image
Time cost
Time (min)
6
4
2
0
↓ 4.8%
Point cloud
Voxel
Our image
(b)
Equiformer
↓11.05%
↓7.67%
SchNet
↓2.97%
↓1.61%
12.5
13
13.5
14
Average MAE on QM9
Baseline
EDG
EDG++
(d)